Step 1: Brainstormed Problem Ideas (Shared Board)

1. **Recipe Meal Planner:** Users struggle to plan weekly meals based on ingredients they already have, leading to food waste.

2. **Flight Price Tracker:** Travelers miss out on cheap flights because tracking fare fluctuations across platforms is tedious.

3. **Public Transit Delay Predictor:** Commuters lack real-time insights on potential delay patterns during peak hours.

4. **Local Event Finder:** Residents and tourists miss small, local cultural events due to scattered event listings.

5. **Personal Finance Budget Categorizer:** Small business owners struggle to automatically tag and categorize raw transaction logs.

6. **Crypto & Stock Alerts:** Retail investors miss sudden volatility triggers on niche assets.

7. **Movie & TV Recommendation Engine:** Viewers experience "decision fatigue" when choosing what to watch across multiple streaming services.

8. **Fitness & Nutrition Tracker:** Users struggle to calculate accurate micro/macronutrient totals for custom home-cooked recipes.

9. **Weather-Based Activity Planner:** Outdoor enthusiasts have plans ruined by hyper-local, sudden weather changes.

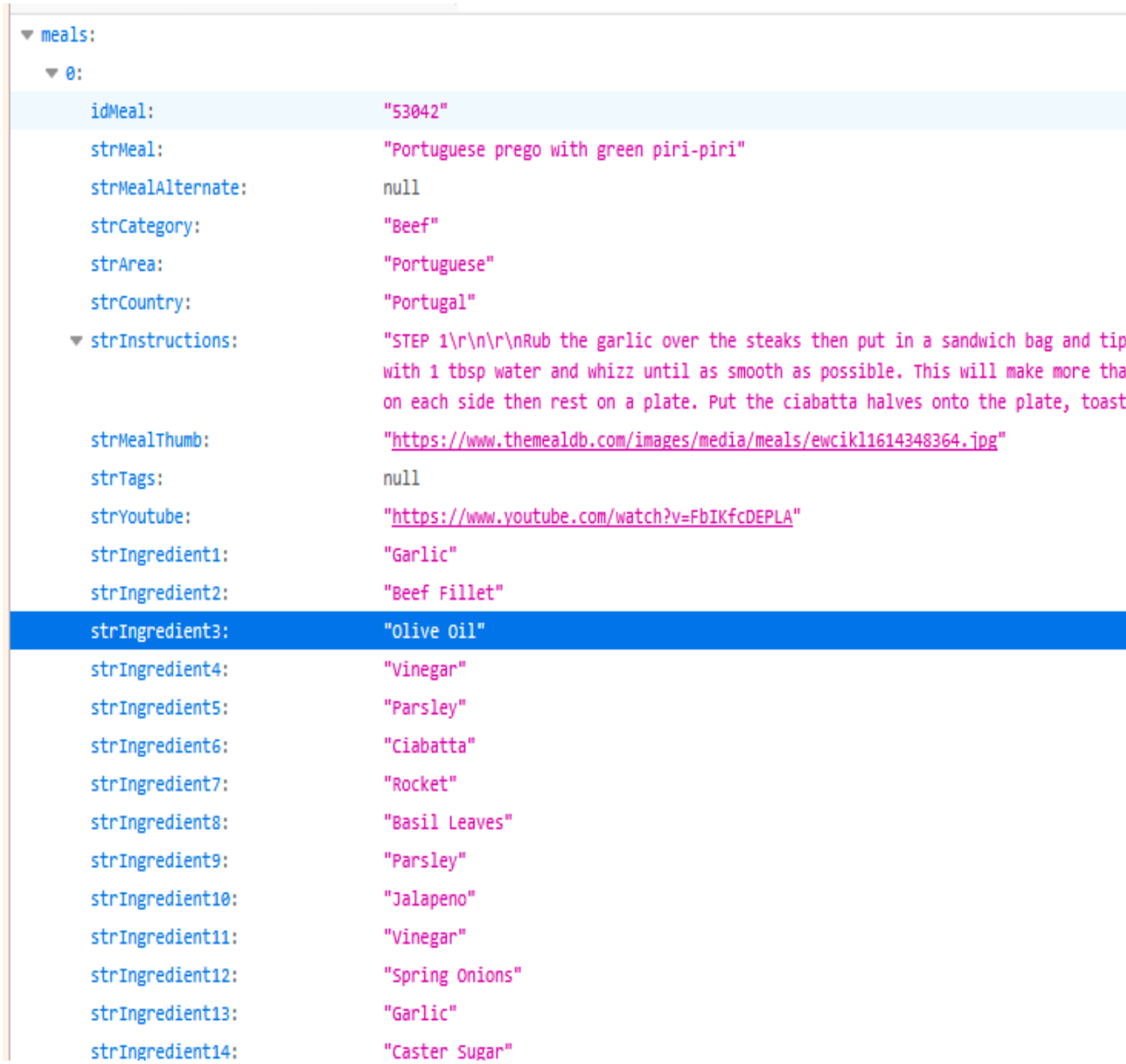
10. **Book Recommendation & Reading Goal Tracker:** Readers struggle to discover books tailored to specific niches or reading speed goals.

2. API Candidate Exploration & Test Requests

API 1: TheMealDB API (For meal ingredients and recipes)

- **Test Request:** https://www.themealdb.com/api/json/v1/1/lookup.php?i=53042

- **JSON Response:**




```
meals:
  0:
    idMeal:            "53042"
    strMeal:           "Portuguese prego with green piri-piri"
    strMealAlternate:  null
    strCategory:       "Beef"
    strArea:           "Portuguese"
    strCountry:        "Portugal"
    strInstructions:   "STEP 1\r\n\r\nRub the garlic over the steaks then put in a sandwich bag and tip
                       with 1 tbsp water and whizz until as smooth as possible. This will make more tha
                       on each side then rest on a plate. Put the ciabatta halves onto the plate, toast
    strMealThumb:      "https://www.themealdb.com/images/media/meals/ewcikl1614348364.jpg"
    strTags:           null
    strYoutube:        "https://www.youtube.com/watch?v=FbIKfcDEPLA"
    strIngredient1:    "Garlic"
    strIngredient2:    "Beef Fillet"
    strIngredient3:    "Olive Oil"
    strIngredient4:    "Vinegar"
    strIngredient5:    "Parsley"
    strIngredient6:    "Ciabatta"
    strIngredient7:    "Rocket"
    strIngredient8:    "Basil Leaves"
    strIngredient9:    "Parsley"
    strIngredient10:   "Jalapeno"
    strIngredient11:   "Vinegar"
    strIngredient12:   "Spring Onions"
    strIngredient13:   "Garlic"
    strIngredient14:   "Caster Sugar"
```


.....

API 2: Open Library API (Api for Literacy)

- **Test Request:** https://openlibrary.org/search.json?q=the+lord+of+the+rings

- **JSON Response:**

numFound	1106
start	0
numFoundExact	true
num_found	1106
documentation_url	"https://openlibrary.org/dev/docs/api/search"
q	"the lord of the rings"
offset	null
docs	
0	
author_key	
0	"OL26320A"
author_name	
0	"J.R.R. Tolkien"
cover_edition_key	"OL51694024M"
cover_height	731
cover_i	14625765
cover_width	394
ebook_access	"borrowable"
edition_count	253
first_publish_year	1954
has_fulltext	true

.....

API 3: Open-Weather API

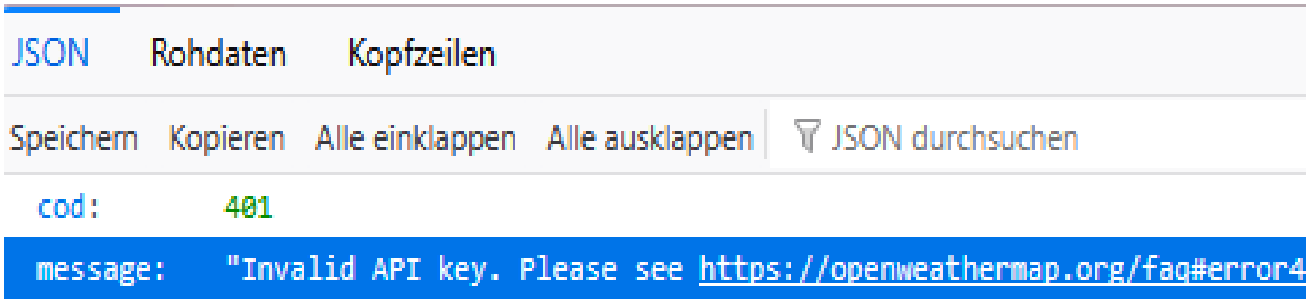
Test Request:

https://api.openweathermap.org/data/2.5/weather?lat={lat}&lon={lon}&appid={API%20key}

JSON Response:


```
 {
  "coord": { "lon": 8.2711, "lat": 50.0000 },
  "weather": [
    { "id": 800, "main": "Clear", "description":
"clear sky", "icon": "01d" }
  ],
  "main": {
    "temp": 18.5,
    "feels_like": 18.1,
    "temp_min": 16.0,
    "temp_max": 20.0,
    "humidity": 55
  },
  "name": "Mainz",
  "cod": 200
```

.....


JSON
Rohdaten
Kopfzeilen
Speichern
Kopieren
Alle einklappen
Alle ausklappen
JSON durchsuchen
cod: 401
message: "Invalid API key. Please see https://openweathermap.org/faq#error4

---

3. Selected Idea: Open Library API

- **Target User:** Voracious readers, students, and book club members who struggle to find specific book metadata and want to track their annual reading goals based on page counts and genres.
- **The Problem:** Readers often experience "decision fatigue" when looking for their next book. Additionally, tracking reading progress manually across different editions (which have varying page counts) is tedious and often inaccurate.
- **Business Value:**
    - **Direct Value Proposition:** Provides a centralized, automated tracker that pulls precise book data (like exact page counts per edition), keeping users motivated to hit their reading targets.
    - **Monetization & Market Potential:** High potential for targeted affiliate marketing (linking directly to bookstore purchases), a premium subscription model offering advanced reading analytics, and potential B2B partnerships with local libraries or independent bookstores.